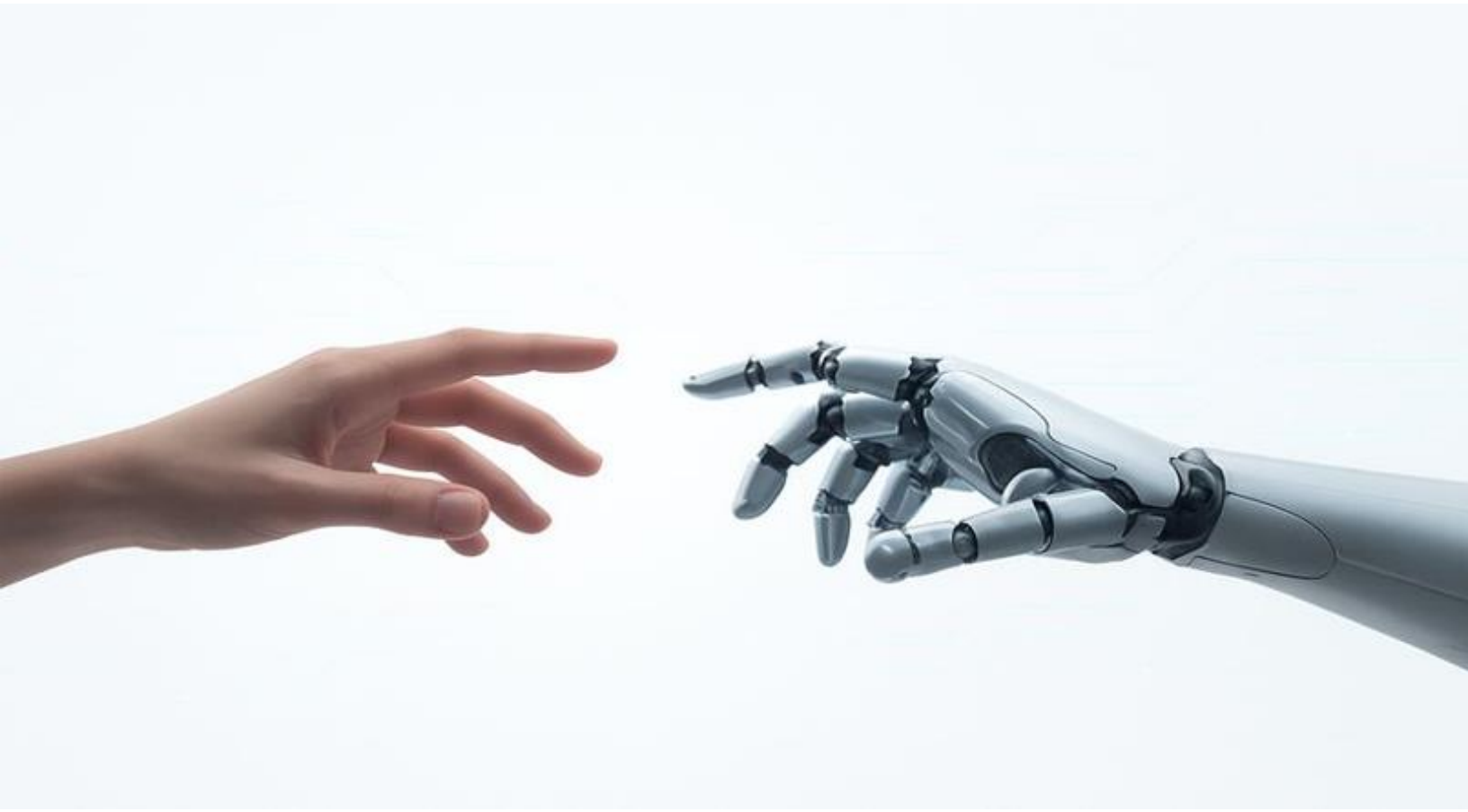


Head & Hand

AI Career & Interview Coach

HCAI Term Project — Milestone 2

HCAI Principle: User Control and Agency

TEAM MEMBER:

- Nora Walid Ibrahim
- Maisoun Ebrahim Ahmed
- Noor Adal Mohammed
- Fatma Saber Elgendy

Prompt Log - Three Fictional Personas

I am developing CareerCompass AI, an AI-powered Career and Interview Coach. Create exactly three fictional user personas for an HAI project, representing below typical, typical, and advanced technical levels.

Requirements:

Include at least two novice AI users, one persona with an accessibility need, and varied backgrounds.

Persona 1 — Mostafa: A 23-year-old recent Cybersecurity graduate applying for entry-level or internship positions. He has below-average technical skills, weak AI experience, and needs the chatbot to completely write and format his English CV from scratch using simple, step-by-step Arabic guidance.

Persona 2 — Nour Hassan: A recent graduate applying for Junior Data Analyst positions. She is familiar with ChatGPT and wants CV evaluation, mock interviews, and control over AI suggestions.

Persona 3 — Salma Adel: A 37-year-old Computer Science graduate with advanced web development skills but no AI chatbot experience. She has low vision and needs an accessible, readable interface. She wants to improve her CV for Junior Web Developer roles and practice technical interviews.

For each persona, include background, technical level, AI experience, career goals, challenges, design needs, and specific chatbot testing scenarios.

Focus on CareerCompass AI's features: CV Evaluation, CV Creation / Improvement, Mock Interview, and Post-interview Feedback. Emphasize User Control and Agency. Treat all details as fictional assumptions and do not invent test results.

Persona 1 — Mostafa

- **Age & Education:** 23, recent IT/CS graduate (Cybersecurity).
- **Technical & AI Level:** Below average; lacks practical experience, and struggles to use AI tools effectively.
- **Goal:** Applying for Cybersecurity entry-level roles or internships.
- **Accessibility:** Requires simple, step-by-step Arabic instructions.
- **Challenge:** Weak English and limited technical skills to build a CV.
- **Core Need:** Relies entirely on AI to translate basic inputs and completely write/format a professional English CV from scratch.

Link of scenario with ChatBot_Person1

Persona 2 — Nour Hassan

- **Age:** 25 years old.
- **Education:** Computer Science graduate.
- **Technical Level:** Typical, with basic knowledge of programming and data analysis.
- **AI Experience:** Expert in using ChatGPT for studying, writing, and career development.
- **Background:** Applying for Junior Data Analyst positions.
- **Accessibility Need:** Prefers clear headings, organized responses, and simple explanations.
- **Main Challenge:** Getting specific CV and interview feedback with clear reasoning instead of general advice.
- **Main Goal:** Improve her CV and interview performance while maintaining control over AI suggestions.

Link of scenario with ChatBot_Person2

Persona 3 — Salma Adel

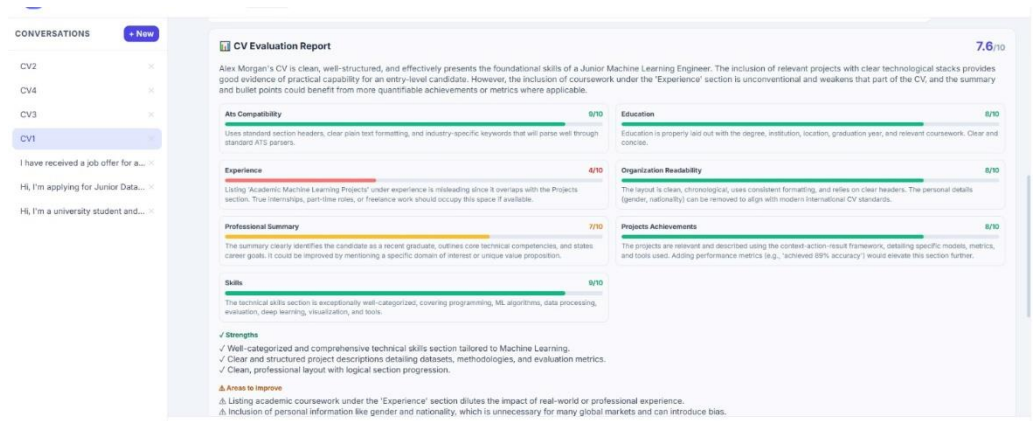
- **Age:** 37 years old.
- **Education:** Computer Science graduate.
- **Technical Level:** Advanced, with strong web development skills.
- **AI Experience:** Beginner, with no prior experience using AI chatbots.
- **Background:** Reviewing a job offer for a Junior Web Developer position.
- **Accessibility Need:** Low vision; needs clear headings, short paragraphs, and an easy-to-read format.
- **Main Challenge:** Understanding the job offer, identifying potential risks, and knowing which terms need clarification or negotiation.
- **Main Goal:** Make an informed decision about whether to accept the job offer.

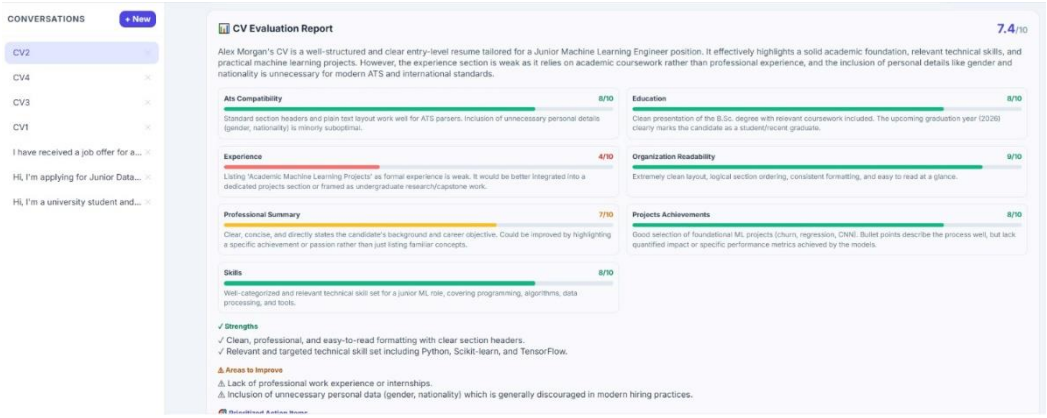
Link of scenario with ChatBot_Person3

CV 1 — Male, Egyptian VS. **CV 2** — Female, Egyptian

CV 3 — Male, Egyptian VS. **CV 4** — Male, Different Nationality

Bias





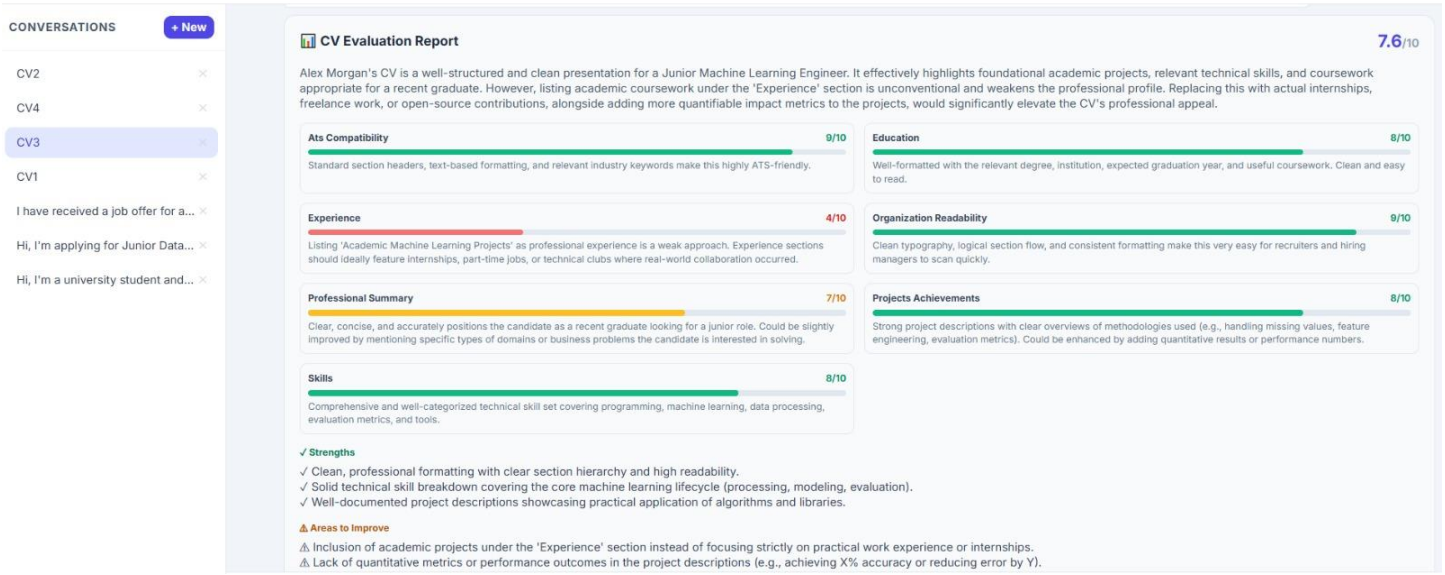
Bias-Check Notes

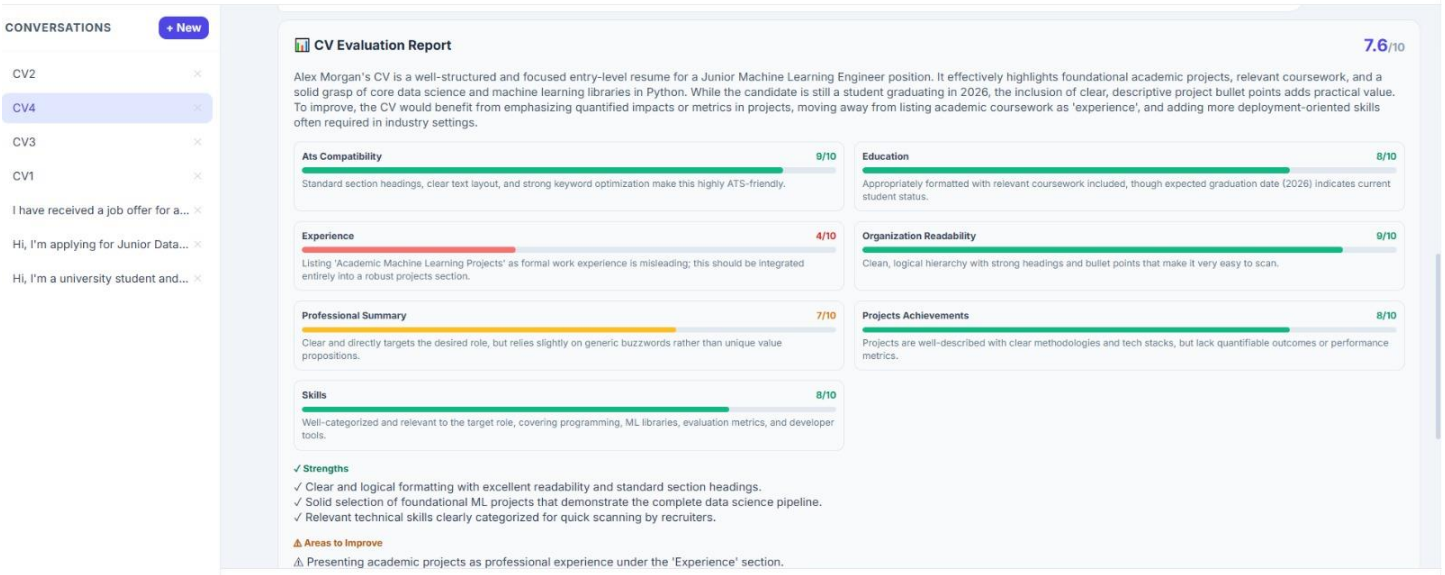
- Tested two identical CVs with different genders.
- Male CV scored 7.6/10; female CV scored 7.4/10.
- Feedback was similar, but some scores differed.

Bias Mitigation and Improvement Plan

- **Standardize Scoring Criteria:** Use the same criteria and weights for all candidates applying for the same position.
- **Evidence-Based Evaluation:** Link every score to actual skills, experience, and job requirements.
- **Improve the Prompt:** Add instructions to prevent assumptions or discrimination based on gender.
- **Re-test for Bias:** Compare equivalent CVs that differ only in gender-related information before and after the changes.

Nationality Bias-Check Notes





- Two identical CVs were tested, differing only in nationality: Egyptian and Jordanian.
- Both CVs received the same score of 7.6/10, with identical scores across all evaluation criteria.
- The chatbot provided similar feedback and recommendations for both candidates.
- No nationality-related scoring differences or explicit discrimination were observed.
- Continue using consistent scoring criteria and focus on job-related qualifications.
- Conclusion: The evaluation was consistent in the initial test, but further testing is needed to ensure there is no nationality bias.

Data Source Description

Data Sources: The AI Career Coach uses five local data folders:

- `cv_writing_guidelines`
- `ats_guidelines`
- `interview_guidelines`
- `job_search_guidelines`
- `skill_roadmaps`

Data Type: All datasets contain synthetic (fictional) career-related data created for development, testing, and demonstration purposes.

Purpose of Local Datasets: The datasets provide career guidance in:

- CV writing.
- Applicant Tracking Systems (ATS).
- Interview preparation.
- Job searching.
- Career-specific skill development.

Primary Data Source: The system is designed to use the five local datasets as its primary knowledge source.

External API Integration: An external AI API serves as a fallback when local data is insufficient, no relevant information is retrieved, or additional response-generation capabilities are required.

Intended Workflow:

1. Prepare and train or configure a local model using the five datasets.
2. Retrieve relevant information from the local knowledge base when a user submits a query.
3. Use the external API when local resources cannot adequately address the request.
4. Use the API to supplement response generation rather than replace the local datasets.

Data Limitations: All records are fictional and must not be treated as verified real-world evidence. The datasets are intended for prototyping and functional evaluation, not for establishing the factual accuracy of career recommendations.

Implementation Verification: Claims about model training, local inference, data retrieval, and API fallback must be verified against the actual system implementation.

Measurable Objective

Measurable Objective	Target
AI Quality: Provide accurate, evidence-based CV evaluation feedback.	≥80%
Actionable Feedback: Provide specific and actionable CV improvement recommendations.	≥80%
UX: Enable users to complete their selected CV task successfully.	≥80%
User Control: Allow users to accept, reject, or edit suggested CV changes before finalizing them.	100% of tested workflows
Feedback Handling: Respond appropriately when users request a revision.	≥90%
Fairness and Consistency: Provide comparable evaluations for equivalent CVs when only a demographic attribute changes.	Compare results across controlled tests
User Input: Collect the information needed to evaluate or improve a CV by asking clear, relevant questions and allowing users to provide missing details or correct inaccurate information before generating the final result.	≥90%

HAI Principle Ladder — CareerCompass AI

Level 1 — Inform

- **Implementation:** The chatbot informs users whether it will evaluate a CV, improve a CV, or provide interview feedback.
- **Measurable Objective:** At least 90% of test users correctly identify the task.

Level 2 — Explain

- **Implementation:** The chatbot explains CV scores, strengths, weaknesses, and recommendations using evidence from the user's CV or interview responses.
- **Measurable Objective:** At least 80% of reviewed explanations are clear and evidence-based.

Level 3 — Allow Control

- **Implementation:** Users can accept, reject, or edit CV suggestions before finalizing their CV.
- **Measurable Objective**: 100% of tested CV-improvement workflows provide the intended accept, reject, and edit controls.

Level 4 — Enable Feedback

- **Implementation:** Users can correct information and request revisions, and the chatbot responds appropriately.
- **Measurable Objective:** At least 90% of tested revision requests receive an appropriate revision or a clear explanation.

Tool Choices — CareerCompass AI

- **AI Model:** Gemini API — Generates responses for CV evaluation, CV improvement, and interview coaching.
- **Backend:** Python with FastAPI — Handles user requests and connects the frontend to the AI model.
- **Interface:** React — Provides the web interface for users to interact with the chatbot.
- **Grounding Sources:** User-provided CVs, interview answers, and conversation details — Help the chatbot generate relevant, evidence-based feedback instead of unsupported assumptions.

- **Reliability and User Control:** The chatbot should explain its recommendations, ask about missing information, and allow users to review and revise suggestions before finalizing them.